

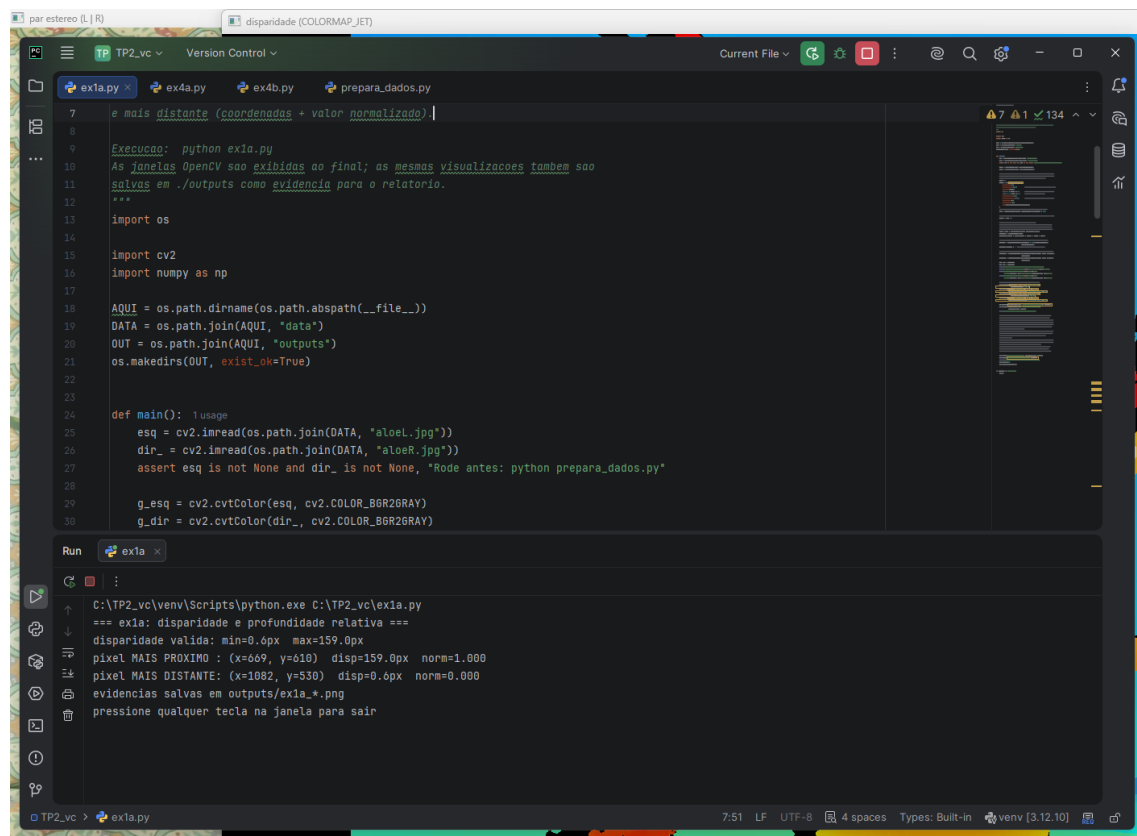
Neste TP foi usado python 3.12 executado em windows, e o comando para reproduzir o ambiente é: Pip install -r requeriments.txt

O treinamento da CNN foi feito em CPU i9-14900kf, assim como a execução dos scripts. Isso é relevante notar pela velocidade de inferência que aparece nos resultados dos scripts.

O script prepara_dados.py gera os dados que foram usados no TP.

Seguem as evidências de execução: (mais evidências na pasta ./outputs)

1 A



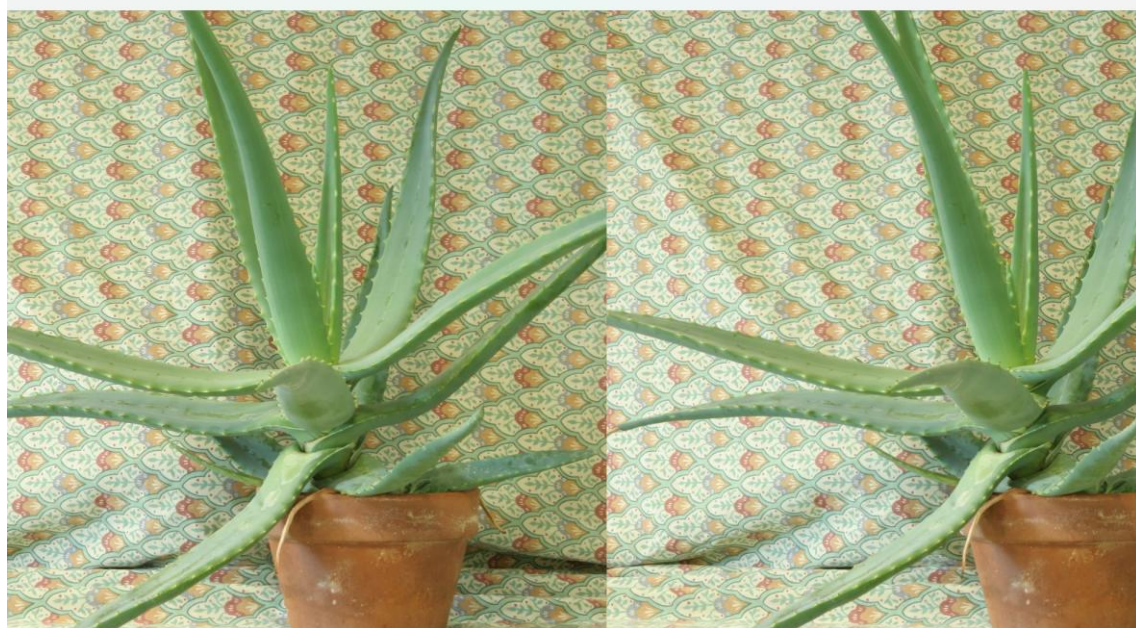
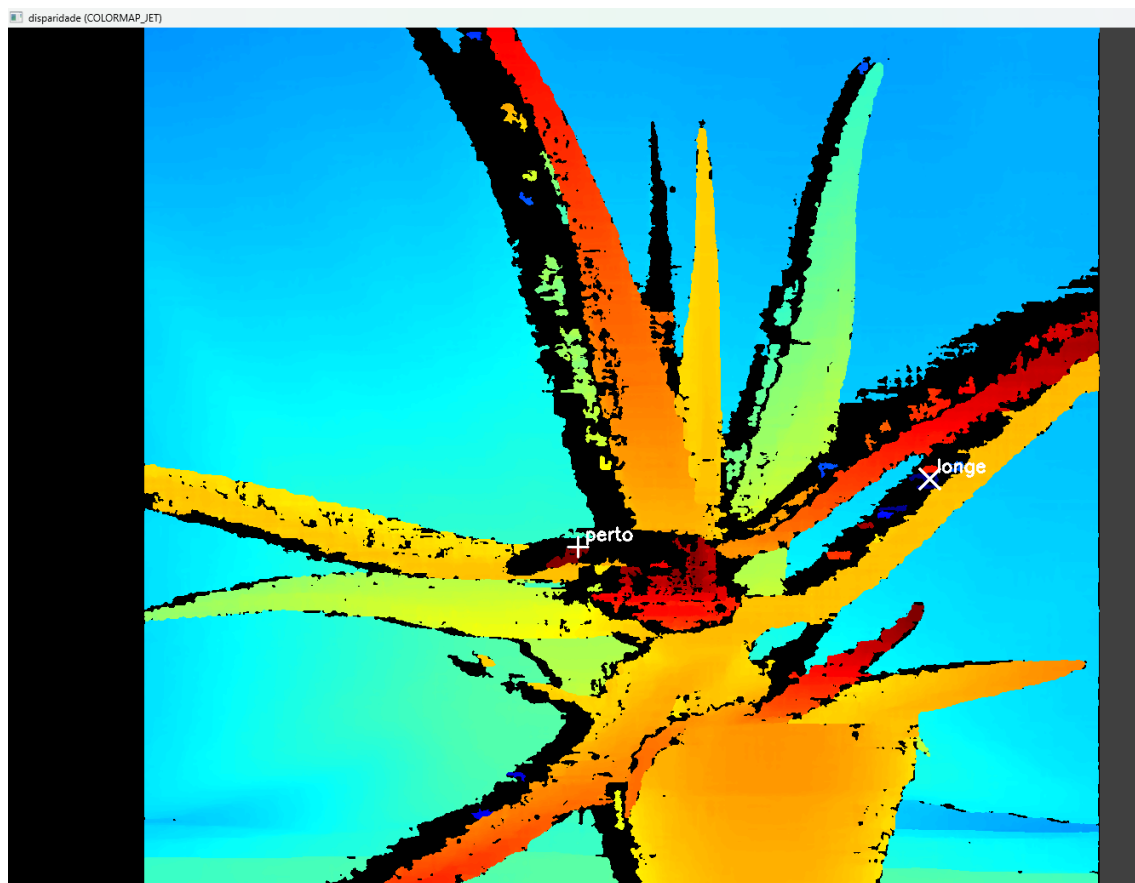
The screenshot shows a Visual Studio Code editor window with a dark theme. The editor is open to a file named `ex1a.py`. The code in the editor is as follows:

```
7 e mais distante (coordenadas + valor normalizado).
8
9 Execução: python ex1a.py
10 As janelas OpenCV são exibidas ao final; as mesmas visualizações também são
11 salvas em ./outputs como evidência para o relatório.
12 ***
13 import os
14
15 import cv2
16 import numpy as np
17
18 AQUI = os.path.dirname(os.path.abspath(__file__))
19 DATA = os.path.join(AQUI, "data")
20 OUT = os.path.join(AQUI, "outputs")
21 os.makedirs(OUT, exist_ok=True)
22
23
24 def main(): 1 usage
25     esq = cv2.imread(os.path.join(DATA, "a1oeL.jpg"))
26     dir_ = cv2.imread(os.path.join(DATA, "a1oeR.jpg"))
27     assert esq is not None and dir_ is not None, "Rode antes: python prepara_dados.py"
28
29     g_esq = cv2.cvtColor(esq, cv2.COLOR_BGR2GRAY)
30     g_dir = cv2.cvtColor(dir_, cv2.COLOR_BGR2GRAY)
```

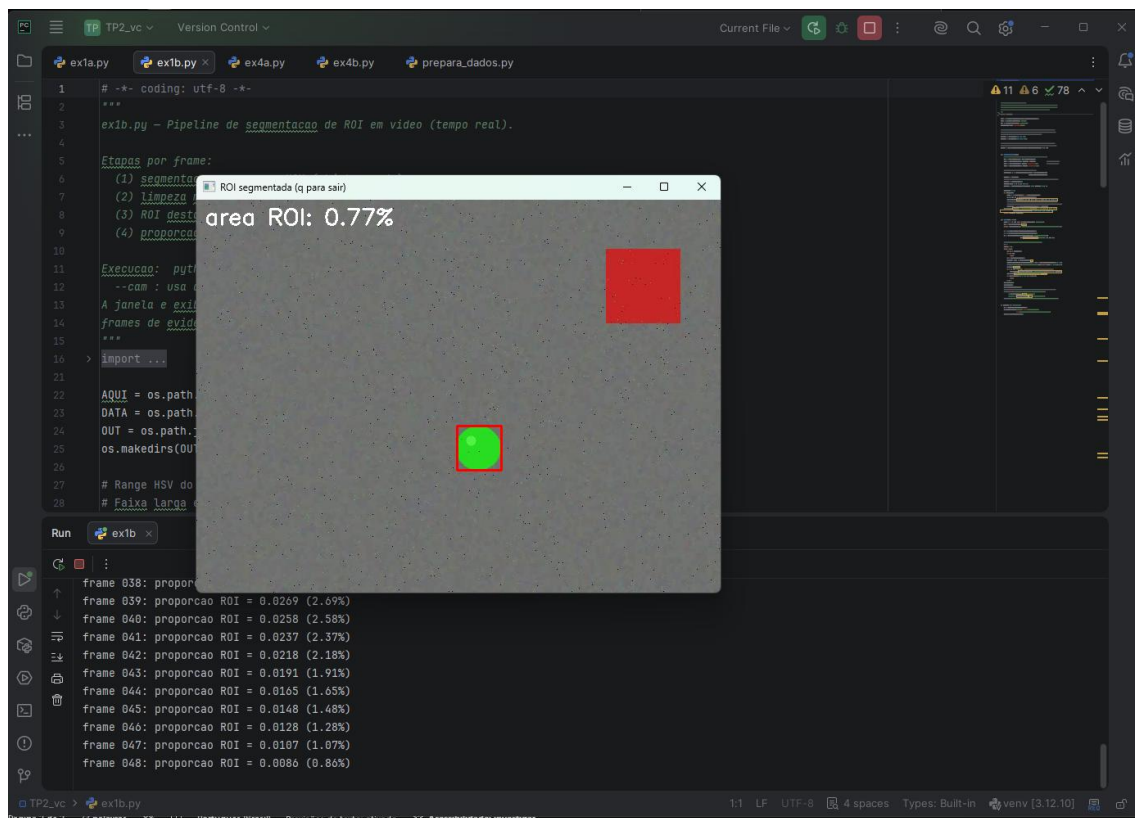
Below the editor, the Run and Debug console shows the output of the script:

```
C:\TP2_vc\venv\Scripts\python.exe C:\TP2_vc\ex1a.py
=== ex1a: disparidade e profundidade relativa ===
disparidade valida: min=0.0px max=159.0px
pixel MAIS PROXIMO : (x=669, y=610) disp=159.0px norm=1.000
pixel MAIS DISTANTE: (x=1082, y=530) disp=0.0px norm=0.000
evidencias salvas em outputs/ex1a*.png
pressione qualquer tecla na janela para sair
```

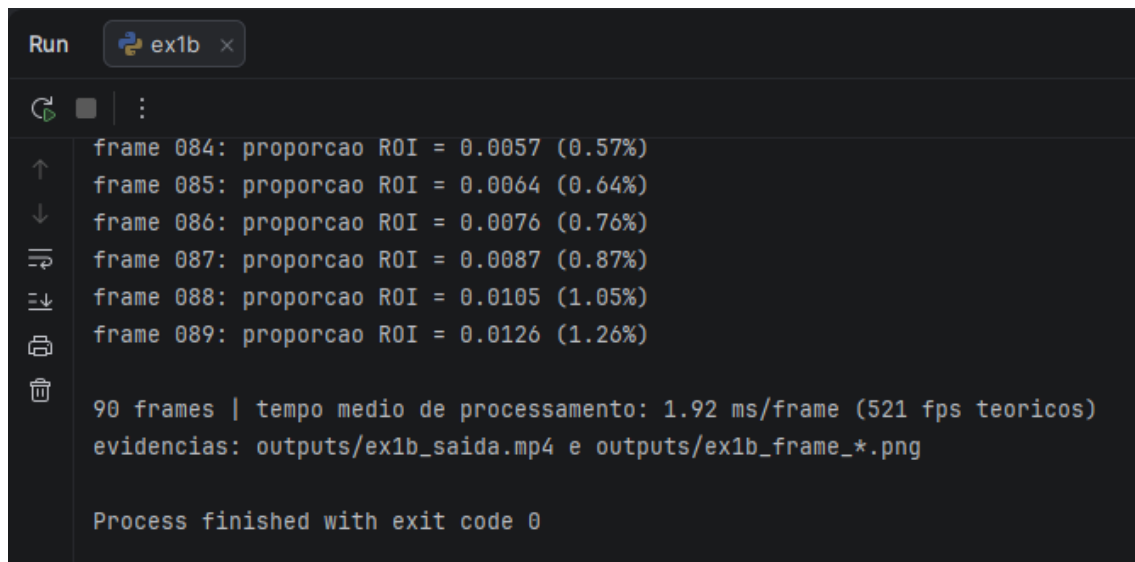
The status bar at the bottom of the editor shows the file is `ex1a.py`, the encoding is `UTF-8`, and the Python environment is `venv [3.12.10]`.



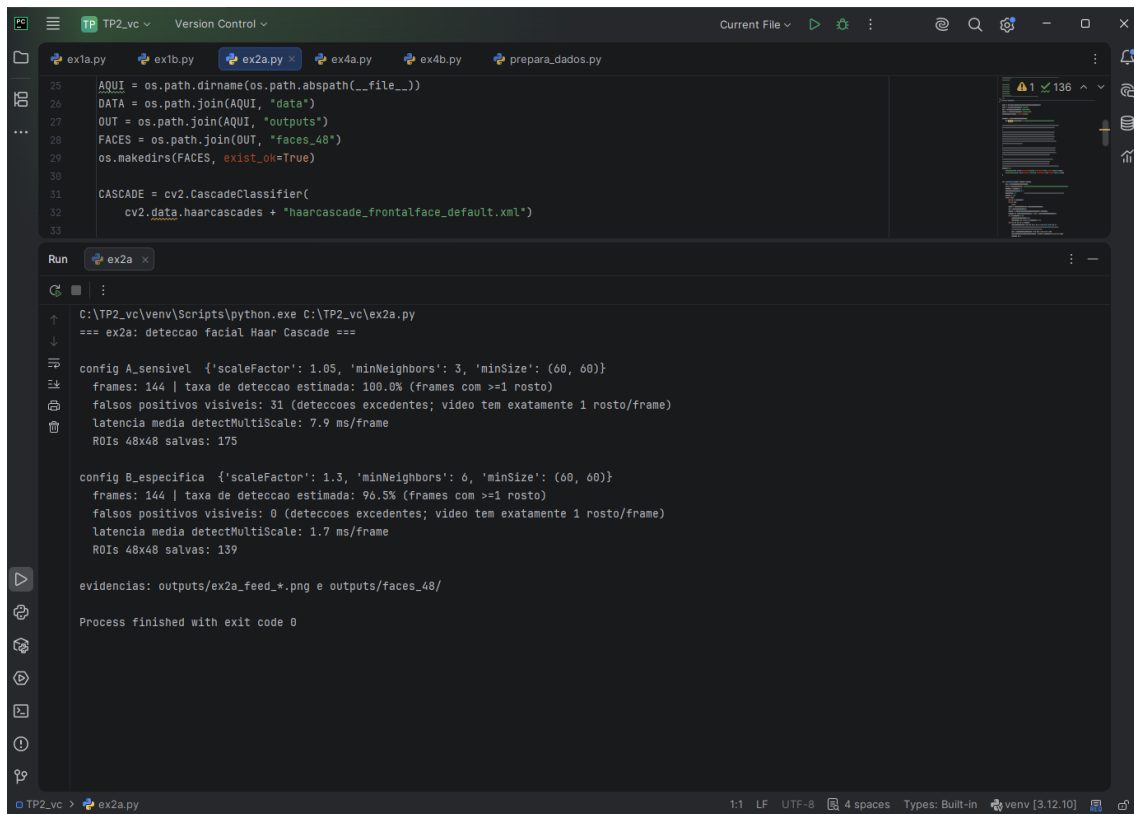
1 B



Saída final do terminal:



2 A – Resultado da comparação no terminal



```
TP2_vc  Version Control  Current File  ex1a.py  ex1b.py  ex2a.py  ex4a.py  ex4b.py  prepara_dados.py
25  AQUI = os.path.dirname(os.path.abspath(__file__))
26  DATA = os.path.join(AQUI, "data")
27  OUT = os.path.join(AQUI, "outputs")
28  FACES = os.path.join(OUT, "faces_48")
29  os.makedirs(FACES, exist_ok=True)
30
31  CASCADE = cv2.CascadeClassifier(
32      cv2.data.haarcascades + "haarcascade_frontalface_default.xml")
33
Run  ex2a
C:\TP2_vc\venv\Scripts\python.exe C:\TP2_vc\ex2a.py
=== ex2a: deteccao facial Haar Cascade ===

config A_sensivel {'scaleFactor': 1.05, 'minNeighbors': 3, 'minSize': (60, 60)}
frames: 144 | taxa de deteccao estimada: 100.0% (frames com >=1 rosto)
falsos positivos visiveis: 31 (deteccoes excedentes; video tem exatamente 1 rosto/frame)
latencia media detectMultiScale: 7.9 ms/frame
ROIs 48x48 salvas: 175

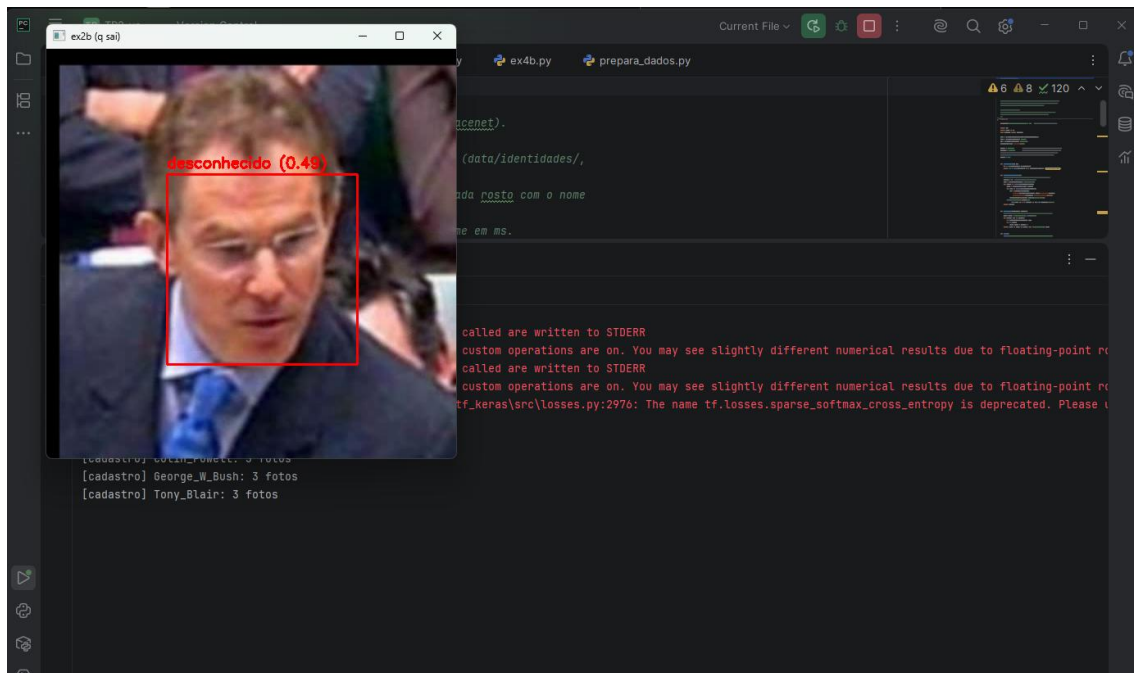
config B_especifica {'scaleFactor': 1.3, 'minNeighbors': 0, 'minSize': (60, 60)}
frames: 144 | taxa de deteccao estimada: 96.5% (frames com >=1 rosto)
falsos positivos visiveis: 0 (deteccoes excedentes; video tem exatamente 1 rosto/frame)
latencia media detectMultiScale: 1.7 ms/frame
ROIs 48x48 salvas: 139

evidencias: outputs/ex2a_feed_*.png e outputs/faces_48/

Process finished with exit code 0
```

2 B –

Durante a execução



Ao final:

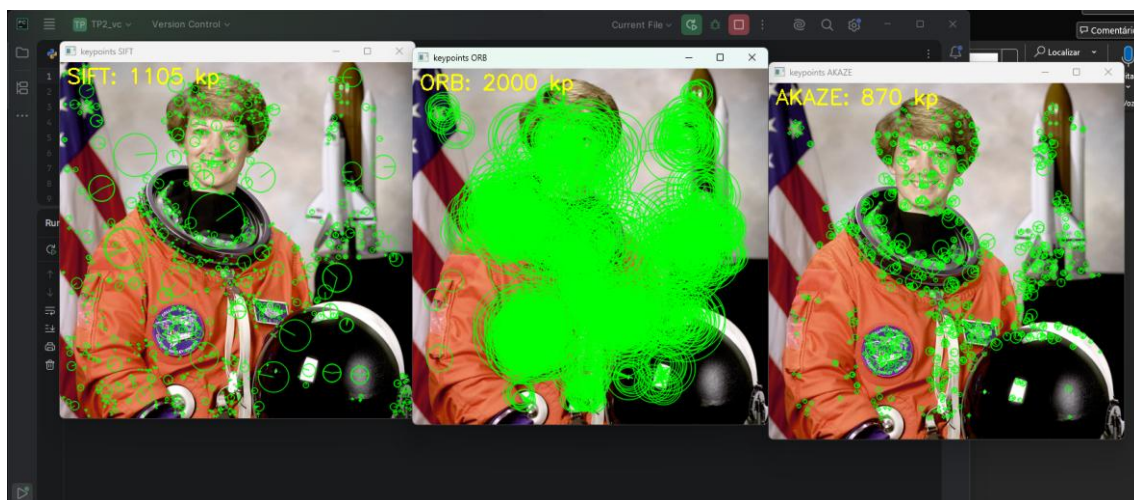

```
Run ex2b x
C:\TP2_vc\venv\Scripts\python.exe C:\TP2_vc\ex2b.py
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1787613666.830815 20920 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point r
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1787613667.700304 20920 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point r
WARNING:tensorflow:From C:\TP2_vc\venv\Lib\site-packages\tf_keras\src\losses.py:2976: The name tf.losses.sparse_softmax_cross_entropy is deprecated. Please u

=== ex2b: reconhecimento facial (DeepFace/Facenet) ===
[cadastro] Colin_Powell: 3 fotos
[cadastro] George_W_Bush: 3 fotos
[cadastro] Tony_Blair: 3 fotos

frames processados: 144
tempo MEDIO de inferencia por frame: 132.5 ms (mediana 129.4 ms, ~7.5 fps)
rotulos atribuidos: {'Colin_Powell': 41, 'desconhecido': 52, 'George_W_Bush': 30, 'Tony_Blair': 33}
evidencias: outputs/ex2b_saida.mp4 e outputs/ex2b_frame_*.png

Process finished with exit code 0
```

3A-

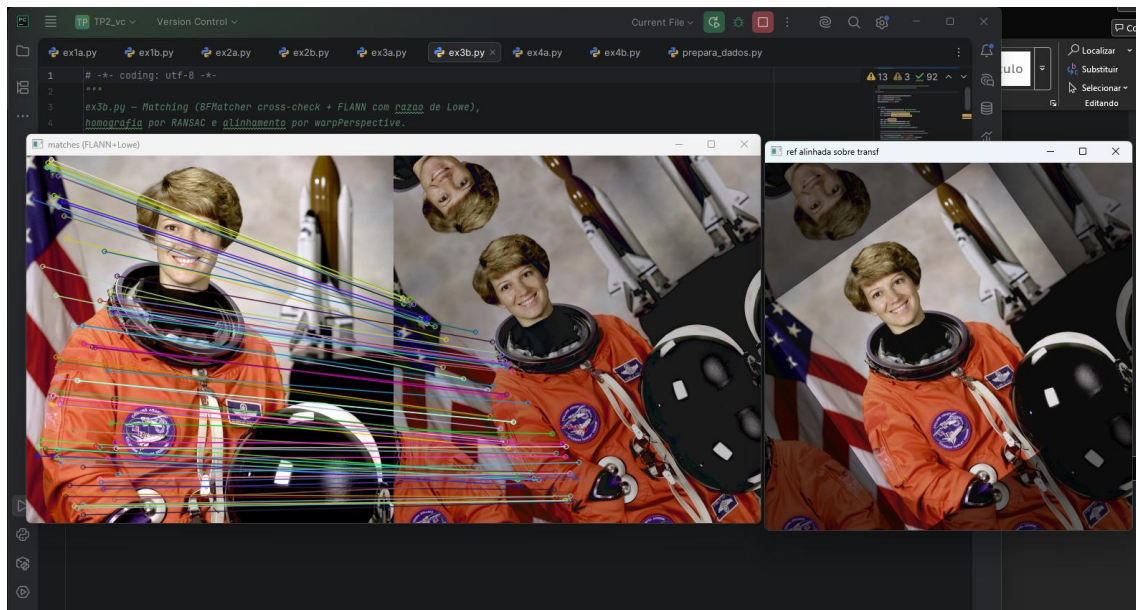


```
Run ex3a x
C:\TP2_vc\venv\Scripts\python.exe C:\TP2_vc\ex3a.py
=== ex3a: comparacao de descritores ===

metodo  kp_ref  kp_transf  tempo_ms  dim  bytes/pt  tipo
SIFT    1105    1187      21.3     128    512  float32
ORB     2000    2000      47.9     32     32   uint8
AKAZE   870     738       14.6     61     61   uint8

evidencias: outputs/ex3a_keypoints_*.png
pressione tecla em uma janela para sair
```

3 B-



```
Run ex3b x
C:\TP2_vc\venv\Scripts\python.exe C:\TP2_vc\ex3b.py
=== ex3b: matching + homografia ===
keypoints: ref=1105 transf=1187
BFMatcher (cross-check): 558 matches
FLANN + razao de Lowe (0.75): 448 matches bons de 1105 candidatos
RANSAC: 425 inliers de 448 matches (94.9%)
homografia estimada:
[[ 0.614  0.4296 -11.3276]
 [-0.4302  0.6137 209.0988]
 [-0.    -0.    1.   ]]
evidencias: outputs/ex3b_matches.png, ex3b_alinhada.png, ex3b_sobreposicao.png
pressione tecla em uma janela para sair

Process finished with exit code 0
|
```

4 A – Foi usada uma rede pequena devido ao processamento em CPU. Segue a estrutura:

The screenshot shows a code editor with a file explorer on the left and a terminal at the bottom. The file explorer shows several Python files: ex1a.py, ex1b.py, ex2a.py, ex2b.py, ex3a.py, ex3b.py, ex4a.py (selected), ex4b.py, and prepara_dados.py. The code in ex4a.py is a Keras CNN model for Fashion-MNIST classification. The terminal output shows the execution of the script, including a warning about logging, the model definition, and a summary of the model layers and parameters.

```
1 #-*- coding: utf-8 -*-
2 ***
3 ex4a.py - CNN do zero para classificacao do Fashion-MNIST (10 classes).
4
5 Conforme instrucoes do trabalho: dataset Fashion-MNIST e Keras rodando com
6 BACKEND TENSORFLOW (definido explicitamente via KERAS_BACKEND antes do
7 import). Arquitetura: 2 blocos Conv2D+MaxPooling -> Flatten -> Dense ->
8 softmax. Treino de 10 epochs com curvas de acuracia/loss e acuracia de teste.
9
```

Run ex4a

```
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
10000 00:00:1787614132.969926 3216 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point p
=== ex4a: CNN Fashion-MNIST (backend: tensorflow) ===
Model: "cnn_fmnist"
```

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 28, 28, 32)	320
max_pooling2d (MaxPooling2D)	(None, 14, 14, 32)	0
conv2d_1 (Conv2D)	(None, 14, 14, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 7, 7, 64)	0
Flatten (Flatten)	(None, 3136)	0
features (Dense)	(None, 128)	401,536
dropout (Dropout)	(None, 128)	0
dense (Dense)	(None, 10)	1,290

Total params: 421,642 (1.61 MB)
Trainable params: 421,642 (1.61 MB)
Non-trainable params: 0 (0.00 B)

Process finished with exit code 0

Resultado final

The screenshot shows a terminal window with the output of the training process. The output displays the accuracy and loss for each epoch, as well as the final accuracy and loss for the test set. The final accuracy is 0.9160 and the final loss is 0.2392.

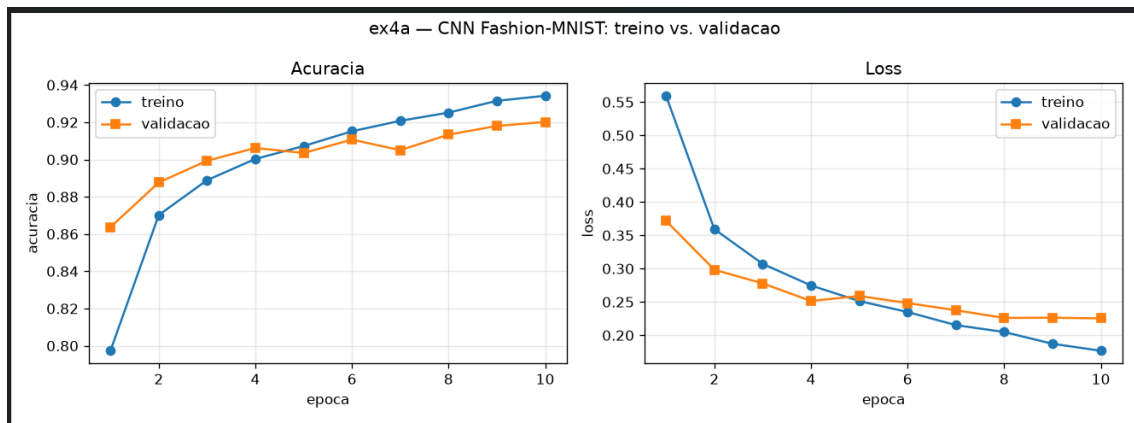
```
Run ex4a
```

```
422/422 - 3s - 6ms/step - accuracy: 0.9151 - loss: 0.2349 - val_accuracy: 0.9107 - val_loss: 0.2483
Epoch 7/10
422/422 - 3s - 6ms/step - accuracy: 0.9208 - loss: 0.2152 - val_accuracy: 0.9050 - val_loss: 0.2375
Epoch 8/10
422/422 - 3s - 6ms/step - accuracy: 0.9252 - loss: 0.2049 - val_accuracy: 0.9133 - val_loss: 0.2260
Epoch 9/10
422/422 - 3s - 6ms/step - accuracy: 0.9315 - loss: 0.1872 - val_accuracy: 0.9180 - val_loss: 0.2263
Epoch 10/10
422/422 - 3s - 6ms/step - accuracy: 0.9342 - loss: 0.1767 - val_accuracy: 0.9202 - val_loss: 0.2253

acuracia no conjunto de TESTE: 0.9160 (loss 0.2392)
grafico salvo em outputs/ex4a_curvas.png
gap final treino-validacao: acc=+0.0140 loss=+0.0486
modelo salvo em outputs/modelo_fmnist.keras

Process finished with exit code 0
```

Curvas

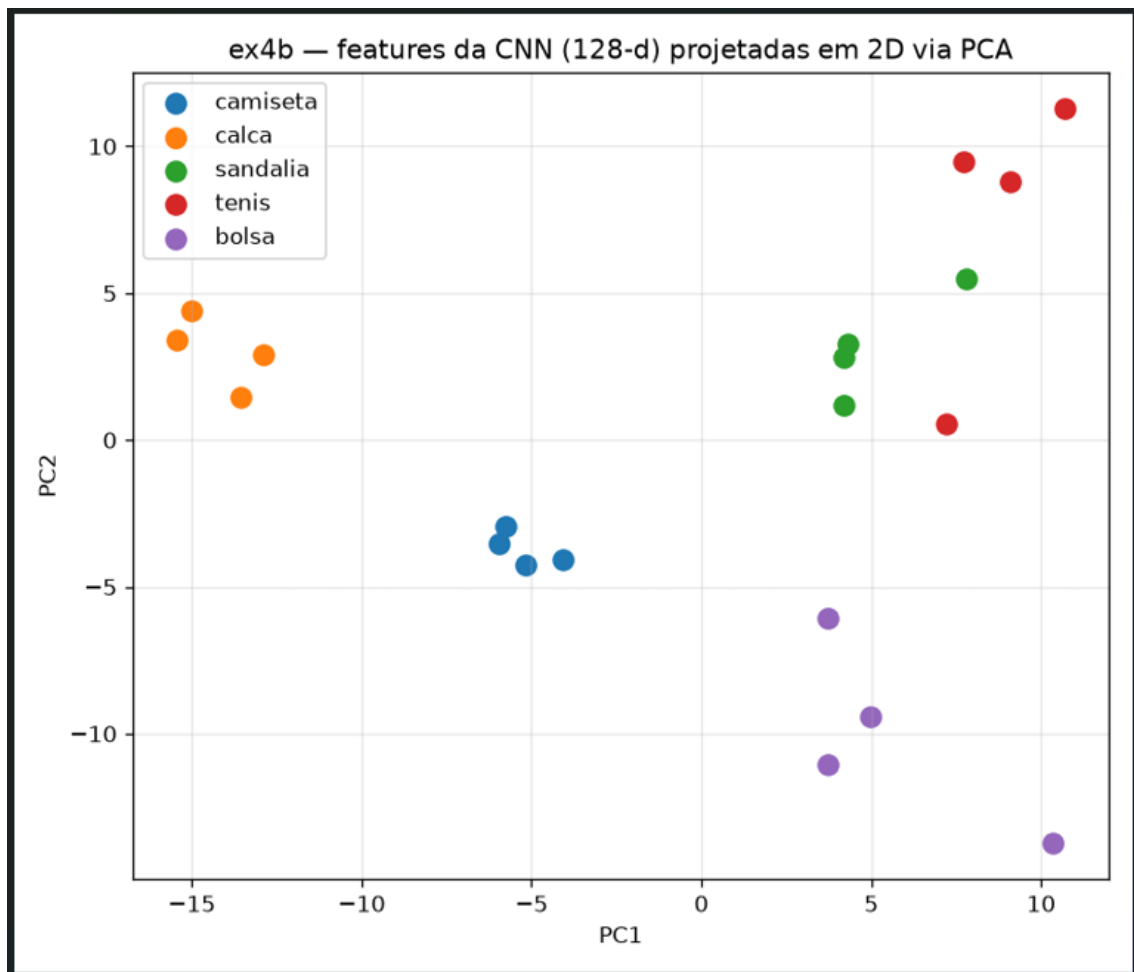


4 B-

```
Run ex4b x
C:\TP2_vc\venv\Scripts\python.exe C:\TP2_vc\ex4b.py
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1787614810.166891 5856 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results du
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1787614811.077788 5856 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results du
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed,
=== ex4b: features CNN + PCA ===
features extraidas: (20, 128) (20 imagens x 128-d)
variancia explicada pelos 2 PCs: 60.9%
scatter salvo em outputs/ex4b_pca_scatter.png
ORB nas mesmas imagens: keypoints por imagem = [13, 0, 15, 0, 6, 17, 46, 13, 40, 48, 46, 46, 40, 31, 30, 45, 0, 21, 14, 12]
media 24.1 kp/imagem; 3 imagens sem nenhum kp

Process finished with exit code 0
```

PCA (2 componentes)



Apesar de a variância explicada não ser tão alta, considerando a redução de dimensionalidade “absurda” do vetor, ainda conseguimos “tênis” e “sandália” próximos, sugerindo um cluster no canto superior direito (a depender do critério, claro). Isso mostra que a última camada densa da rede (antes da softmax) traz distinção do que seria um “sapato”, ou algo parecido com isso, apontando retenção de semântica do objeto. O Orb por sua vez só carrega semelhança geométrica, podendo 2 sapatos diferentes serem encarados como coisas completamente distintas.

Github: [Dev-RMV/TP2_vc](#)